Macroeconomics Notes

Abror Maksudov

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Contents

1 Gross Domestic Product (GDP)

Gross Domestic Product (GDP) is the total market value of all **final goods and services** produced **within a country's borders** in **a specific time period**.

GPD does not include:

- Intermediate goods
- Non-market activities
- Used goods transactions
- Financial transactions
- Illegal activities
- Environmental externalities

2 Methods of Measuring GDP

1. **Production (Value-Added) Approach:**
   - Aggregates the value added at each stage of production.
   - Value added = Revenue − Cost of intermediate inputs.
2. **Expenditure Approach:**
   - Adds up total spending on final goods and services.
   - GDP = Consumption + Investment + Government Spending + (Exports − Imports).
3. **Income Approach:**
   - Sums all incomes earned by households and businesses during production.
   - GDP = Labor income + Capital income.
   - GDP = Wages + Rent + Interest + Profit + . . .

3 Gross National Product (GNP)

Gross National Product (GNP) is the total market value of all **final goods and services** produced **by a country's residents and businesses**, regardless of where that production takes place **during a specific time period**.

- GDP focuses on *where* production happens.
- GNP focuses on *who* produces it (the country's citizens and businesses).

The relationship between GNP and GDP:

$$\text{GNP} = \text{GDP} + \text{Net Factor Income from Abroad}$$

Where "Net Factor Income from Abroad" represents income earned by domestic residents from foreign investments minus income earned by foreign residents from domestic investments. This means if a country earns more from abroad than foreigners earn domestically, GNP will be higher than GDP, and vice versa.

4 National Income Accounting Identity

The relationship between a country's total income and its expenditures:

$$\text{Y} = \text{C} + \text{I} + \text{G} + (\text{X} - \text{M})$$

Where Y is GDP, C is household consumption, I is business investment, G is government purchases, and (X − M) is net exports.

1. **Consumption (C)**
   - *Definition:* All household spending on goods and services, excluding new housing.
   - *Includes:*
     - **Durable Goods:** Long-lasting items (cars, household appliances, furniture).
     - **Non-Durable Goods:** Quick-use items (food, clothing, gasoline).
     - **Services:** Non-physical, intangible products (healthcare, legal services, education).
2. **Investment (I)**
   - *Definition:* Private sector spending on capital goods and changes in inventories.
   - *Includes:*
     - **Business Fixed Investment:** Capital expenditures by firms (equipment, factories).
     - **Residential Fixed Investment:** Spending on new housing units by households and landlords (new home constructions and purchases, major home renovations).
     - **Inventory Investment:** Changes in the stock of unsold goods held by firms.
   - *Excludes:* Financial assets like stocks, bonds, or cryptocurrencies, as these do not represent production.
3. **Government Spending (G)**
   - *Definition:* Government purchases of goods and services.
   - *Includes:*
     - Federal, state, and local spending on goods and services.

- *Excludes:* Transfer payments (social security, unemployment benefits, welfare), because these are not payments for goods or services.

4. **Net Exports (NX = Exports − Imports)**
   - *Definition:* The difference between exports (X) and imports (M).
   - *Includes:*
     - **Exports (X):** Goods and services produced domestically and sold to foreign buyers.
     - **Imports (M):** Goods and services produced abroad and purchased by domestic consumers.
   - Can result in trade surplus (positive) or deficit (negative).

5 Typical GDP Component Distribution

- Consumption (C): $\sim$ 70% – *Largest component*
- Investment (I): $\sim$ 15-20%
- Government (G): $\sim$ 15-20%
- Net Exports (NX): $\sim \pm5\%$ – *Smallest component*